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Multispecies fisheries management and conservation: tactical applications using models of intermediate complexity

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Abstract

Stakeholders increasingly expect ecosystem assessments as part of advice on fisheries management. Quantitative models to support fisheries decision-making may be either strategic ('big picture', direction-setting and contextual) or tactical (focused on management actions on short timescales), with some strategic models informing the development of tactical models. We describe and review 'Models of Intermediate Complexity for Ecosystem assessments' (MICE) that have a tactical focus, including use as ecosystem assessment tools. MICE are context- and question-driven and limit complexity by restricting the focus to those components of the ecosystem needed to address the main effects of the management question under consideration. Stakeholder participation and dialogue is an integral part of this process. MICE estimate parameters through fitting to data, use statistical diagnostic tools to evaluate model performance and account for a broad range of uncertainties. These models therefore address many of the impediments to greater use of ecosystem models in strategic and particularly tactical decision-making for marine resource management and conservation. MICE are capable of producing outputs that could be used for tactical decision-making, but our summary of existing models suggests this has not occurred in any meaningful way to date. We use a model of the pelagic ecosystem in the Coral Sea and a linked catchment and ocean model of the Gulf of Carpentaria, Australia, to illustrate how MICE can be constructed. We summarize the major advantages of the approach, indicate opportunities for the development of further applications and identify the major challenges to broad adoption of the approach.

Keywords Coral Sea pelagic fishery, ecosystem approach to fisheries, ecosystem model, minimally realistic model, multispecies, northern prawn fishery

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Introduction

There is a growing desire to extend the tools on which fisheries management advice is based to account for 'ecosystem considerations', and hence achieve an ecosystem approach to fisheries (EAF) (Smith *et al.* 2007). Although the goals of EAF can be accomplished in some cases without the need for models, use of appropriate models is seen by many as key to the implementation of an EAF. There has been considerable recent work to classify and compare the current range of multispecies and ecosystem models, both in a local context (e.g. Hollowed *et al.* 2011) and globally (e.g. Hollowed *et al.* 2000; Fulton *et al.* 2003; Plagányi 2007; FAO 2008). These models may be categorized as either (i) conceptual (aimed at developing an understanding of ecosystem processes); (ii) strategic (focused on broad scale assessment of directions and patterns of change); or (iii) tactical (directed at supporting specific management decisions) (FAO 2008). While it is generally recognized that most ecosystem models will be used in a conceptual and strategic context (e.g. FAO 2008), actual fisheries management primarily involves tactical decisions that impact day-to-day regulations in fisheries.

Unlike conceptual explorations and strategic planning, which usually relate to broad policy goals (e.g. system-wide goals for exploitation and maintenance of biodiversity), tactical decision-making relates to regulations that often change on an annual basis, such as limits on removals and effort, gear restrictions, and area and seasonal closures. Tactical management advice is based on a set of harvest control rules in an increasing number of jurisdictions, such as the United States, Australia, New Zealand and South Africa. While the specifics of harvest control rules differ among jurisdictions, the information needed on an annual basis to apply them generally include values for reference points and an estimate of current population size – and perhaps its age, length and sex structure – and/or exploitation rate relative to reference points (potentially including economic or social targets, e.g. Rayns 2007). In addition, some control rules account for the estimated uncertainty of these quantities (e.g. NPFMC 2010). Many of the same general principles discussed in this paper apply equally to conventional conservation models such as population viability analysis (PVA), which aim to assess a population's risk of extinction and projected growth under various scenarios.

The models used for tactical advice estimate the values for (some of) their parameters using data collected from the system and apply statistical diagnostic tools (e.g. residual analysis and variance estimation) to evaluate model performance. Basing ecosystem management advice on models that are tied to observational data, and which can be evaluated using standard statistical tools, is one technique for advancing an EAF.

We show that a group of models that we refer to as 'Models of Intermediate Complexity for

Ecosystem assessments' (MICE) satisfy the requirements for tactical decision-making. They can be applied in an ecosystem context and overcome many of the concerns regarding the larger and more complex ecosystem models (e.g. Ecopath and Atlantis) intended for strategic use. MICE are similar to a category of models loosely called minimally realistic models (MRMs). MRMs have the same aim of restricting focus to the limited number of species most likely to have important interactions with a target species (or group of species) of interest (Punt and Butterworth 1995), but MRMs do not explicitly require properties that advance their use as ecosystem assessment tools.

Models of Intermediate Complexity for Ecosystem assessments-type models are beginning to be applied for tactical as well as strategic fisheries decision-making (e.g. Danielsson *et al.* 1997; Plagányi and Butterworth 2012). MICE attempt to incorporate the best features of existing single-species models of relative simplicity and the ability to apply standard statistical methods for parameter estimation, but also take account of broader ecosystem considerations given a few well-defined objectives. As such, these models are more focused than the typical whole-of-ecosystem models such as Atlantis (Fulton *et al.* 2004) and Ecopath with Ecosim (EwE) (Walters *et al.* 1997; Pauly *et al.* 2000). While whole-of-ecosystem models can be applied to only parts of an ecosystem, they tend to be complex and cover substantial portions of an ecosystem. In contrast, MICE only include those components of the ecosystem needed to address the management question under consideration. Thus, MICE are intermediate in complexity between single-species and whole-of-ecosystem models in terms of the number of components (e.g. species or functional groups) or processes represented.

In this paper, we first describe MICE and contrast them with traditional single-species approaches and whole-of-ecosystem models. We use a model of the pelagic ecosystem in the Coral Sea and a prawn-centric model of the Gulf of Carpentaria (GoC), Australia (Fig. 1; Tables 1 and 2), to illustrate how MICE can be constructed. These two examples provide a good contrast in terms of the amount of data available, the structure of the ecosystems and the key questions to be addressed. Important management questions for the Coral Sea pelagic fishery concern the appropriate catch levels and spatial management of three species groups (tuna, sharks and billfish) that are captured during fishing activities. These species share a common prey (squid and myctophids) and are therefore linked so that overfishing of tuna could impact billfish and sharks, including species of conservation concern (Fig. 1a). In contrast, the GoC model couples catchment hydrodynamics (specifically rainfall-runoff) with prawn nursery ground use and offshore prawn dynamics, to answer management questions about the interplay between environmental variability and fishery dynamics and its effect on the recruitment dynamics and sustainability in the northern prawn fishery in the GoC (Fig. 1b).

We then summarize aspects of various existing models that could act as MICE, including several which are currently under development in a management context – highlighting which models conform to the principles of MICE. Finally, we summarize the major advantages of the approach, indicate potential future applications and identify the major challenges to the broad adoption of the approach.

MICE, single-species and whole-of-ecosystem models

Models of Intermediate Complexity for Ecosystem assessments are intermediate in complexity between traditional single-species stock assessments, particularly those based on the integrated analysis paradigm (e.g., Fournier and Archibald 1982; Maunder 2003), and whole-of-ecosystem models (Table 1). In general, MICE attempt to explain the underlying ecological processes for a limited group of populations (typically <10) subject to fishing and anthropogenic interactions (Fig. 1a,b) and include at least one explicit representation of an ecological process (e.g. interspecific interaction or spatial habitat use). The major components of MICE comprise a model of the ecological system, how it is impacted by anthropogenic factors, and how the ecological and human processes represented in the model impact on, and are impacted by, management (Fig. 1). However, in common with whole-of-ecosystem models, and unlike most traditional single-species assessment models, MICE can have linked physical models as well as models of human behaviour. The former models would be used to quantify impacts of environmental forcing, as well as two-way relationships between habitat and resources (e.g. to account for impacts of trawling). The latter elements represent social and economic processes

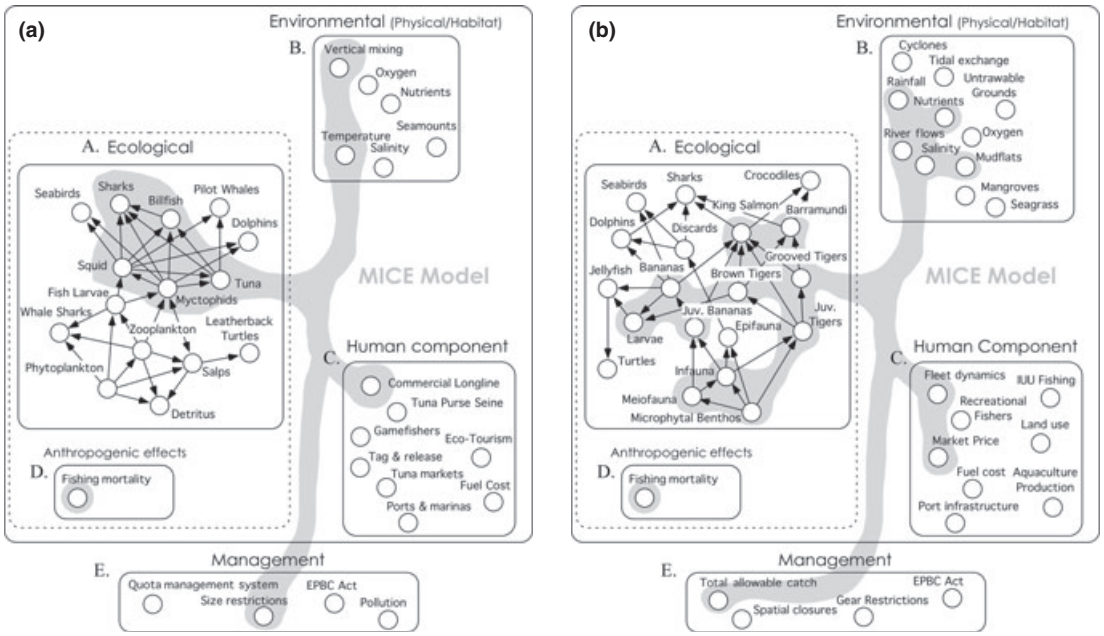


Figure 1 Overview of two Models of Intermediate Complexity for Ecosystem assessments (MICE) examples illustrating (a) the direct and indirect impacts of variable size-specific catches for tuna, sharks and billfish by commercial long-liners in the Coral Sea, and (b) how rainfall and catchment dynamics influence prawn population dynamics and fisheries revenue in the Gulf of Carpentaria, Australia. Inter-connected components are (A) Ecological information, including multispecies trophic or competitive network and spatial habitat processes; (B) Environmental information, including physical and habitat drivers of ecological processes; (C) Human components of the system, including different fishing sectors and their behaviours, other stake-holders and economic drivers; (D) Anthropogenic effects, including fishing mortality; and (E) Management, including input and output controls as well as other environmental policies. A MICE model would include a subset of components (illustrated by shaded regions) that target a specific question/management objective.

that will impact model forecasts and would represent key feedbacks between resources, resource users and stakeholders involved in management. Hence, traditional single-species assessment models typically include ecological and anthropogenic effects (A and E), indicated by the dotted line in Fig. 1. Whole-of-ecosystem models tend to include components (A, B, C and D), while formal management strategy evaluation (MSE) involves all components (i.e. A–E) (Fig. 1).

Traditional single-species assessments, MICE and whole-of-ecosystem models are designed to answer different sets of questions. Most single-species models are designed for tactical issues, while whole-of-ecosystem models are primarily strategic (Table 1). MICE provide the ability to address tactical questions, including estimation of current abundance, current exploitation rate and reference points, while retaining species interactions or other ecological/social/economic system processes. They can also address more ecosystem-related tactical questions such as predicting future population size, yield and the population structure of closed areas. Accounting for more components than single-species models also allows the consequences of, for example, fishers seeking to maximize economic returns on fishing mortality and future abundance, as well as the impact of protection of dependent species on yield (e.g. Maunder *et al.* 2000), to be evaluated.

MICE are designed based on data available to parameterize the model. Some whole-of-ecosystem models (e.g. EwE) can be fit to many of the types of data used in stock assessments (e.g. Gaichas *et al.* 2011). However, this requires exceptional data sets and computation resources and remains the exception rather than the rule.

It is generally quick to make hindcasts using MICE so parameter estimation can be based on standard optimization methods (e.g. using AD Model Builder, Fournier *et al.* 2011). Unlike whole-of-ecosystem models, it is possible to apply standard methods for quantifying parameter uncertainty such as Markov Chain Monte Carlo

Table 1 Features of three classes of ecological models used for an ecosystem approach to fisheries.

	Model type		
	Single-species assessments	Models of Intermediate Complexity for Ecosystem assessments	Whole-of-ecosystem models
Scope	Primarily tactical	Tactical/strategic Focus on a small number of key species/processes/questions of interest	Strategic Designed to address a range of policy-related questions
Complexity/ flexibility	Single-species; no trophic interactions or physical drivers	Limited number of trophic interactions and ecosystem components (<~10) represented Limited number of key physical drivers Processes and components selected given the problem statement and data available Many processes modelled using proxies	Potentially very large number (~100s) of components Detailed representation of biological processes
Temporal resolution	Annual, except for a few exceptions	Selected to match the question being addressed/available data	Usually small time steps to ensure processes are well presented
Accounting for uncertainty	Always fitted to data (maximum likelihood/ Bayesian) Residual analyses (poor patterns lead to model changes) Alternative model structures Simulation evaluation	As for single-species models	Sometimes fitted to data; generally tuned Less focus on addressing residual patterns Alternative model structures
Implementation	Generally packages (Stock Synthesis, FLR, CASAL, MULTIFAN) Training broadly available Most time is associated with model tuning/ residual analysis	Generally case-specific (but general approaches, e.g. GADGET exist) Considerable development time and time for model tuning	Focus predominantly on main drivers of interest Some major packages (Ecopath with Ecosim, Atlantis); opportunities for training exist
Hybrid models	Generally not (although projection models are often linked to assessment models)	Could couple an independent economic/social/physical model with the ecological model as needed	Could couple an independent economic/social/physical model with the ecological model as needed
Model structure	Limited to components related to the stock and fishery being assessed Modern assessment methods allow for flexibility of functional forms for growth, selectivity, catchability, stock-recruitment relationship	Components (ecological, economic and humans) modelled at a variety of levels of resolution	Packages available allow for considerable flexibility in terms of how processes and components are modelled

(MCMC) sampling, inversion of the Hessian matrix or likelihood profiles. This is critical, because obtaining variance estimates and confidence/credible intervals has become increasingly important in many jurisdictions where catch limits are adjusted depending on the estimated extent of uncertainty (e.g. Shertzer *et al.* 2008; NPFMC 2010).

Table 2 Examples of Models of Intermediate Complexity for Ecosystem assessments strategies for encapsulating complexity as simpler representations.

Strategy	Coral Sea pelagic example	Gulf of Carpentaria catchment prawn example
Selection of model compartments	Include only the focus tuna, billfish and shark species, plus the main prey species (squid and myctophids)	Include only the focus banana and tiger prawn species, important prey species in nursery habitat and two important predators (barramundi and threadfin salmon)
Temporal and spatial resolution	Tailored to match scale relevant to major processes and model objectives, for example, do not model finescale changes in primary productivity, but rather integrate effects over larger model scale	Tailored, for example shorter timescale for meiofaunal processes, but annual time step for fish
Model structure	Limited number of trophic interactions and ecosystem components represented. Processes and components selected given the problem statement and data available.	As for Coral Sea model, and also limited number of key physical drivers
Predator-prey interaction	A logistic or normalized sigmoid function, with output range 0–1, and depending on overall prey depletion, is used to represent the dependence of reproduction on prey abundance because it represents a logical and flexible 'scope for growth region'	Consumption not explicitly modelled, but represented using relationships that represent the net overall impact on a predator of a change in its prey
Recruitment drivers	Use fairly simple relationships or rules to modify spatial recruitment patterns, rather than detailed modelling of, for example, physical transport processes	Fairly simple rules based on catchment flow rates to modify recruitment from inshore to offshore areas
Movement	Squid are modelled (in discrete time) using a transition matrix, while tuna are modelled using preference functions where the preference for a given area depends on a variety of key factors	Movement is not modelled explicitly, but simple rules are used to modify the overlaps between different life stages and species
Human dimension	The fishery is decomposed into various fleets (with grouping based on similar gear characteristics and targeting practices) as based on existing stock assessments for the tuna fisheries in the Coral Sea. There is no explicit behavioural modelling of the fleets	Fairly simple rules (rather than a complex bio-economic model) are used to allocate fishing effort among the prawn species
Representation of complex dynamical processes	Avoid this where necessary by modelling an outcome rather than a dynamical process – for example, considering what consumption affects (i.e., survival, reproduction and growth rate) rather than the process of consumption itself	Rainfall (and/or runoff) at a river basin level is used as an input (rather than explicitly modelled using a coupled hydrodynamic model) and simple rules used to represent physical scouring and salinity changes. Field data (M. Duggan, M. Burford and R. Kenyon, unpublished) informed the choice of an inverse dome-shape to represent the relationship between meiofaunal growth rate and salinity

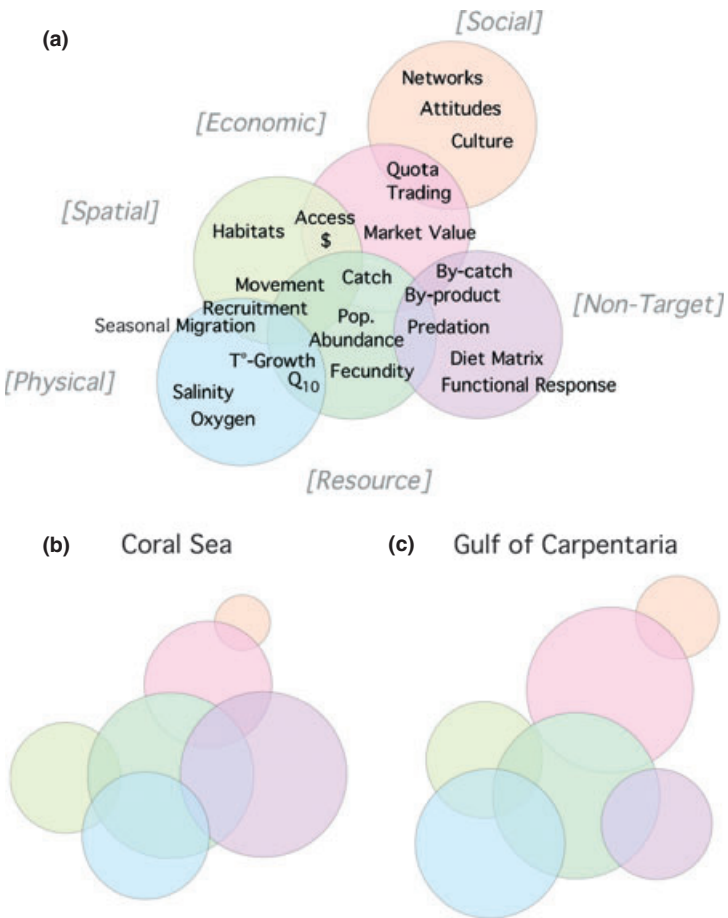


Figure 2 Schematic showing the overlap of data types and information that can be utilized in the parameterization of a Models of Intermediate Complexity for Ecosystem assessments (MICE) model. (a) Broad categories of data include social, economic, spatial (e.g. habitat, heterogeneity), physical and data from resource and non-target species (e.g. predators). The lower plots show the relative amounts of data and overlap for MICE models of (b) the Coral Sea and (c) the Gulf of Carpentaria, Australia, illustrating differences in emphasis on social, economic, spatial, physical and data on resource and non-target species.

Principles and strategies underlying MICE

All models represent subsets of the ecosystem, and care needs to be taken to select an appropriate subset. The structure of each model component is determined by the management questions, availability of data and the degree of similarity in ecological function or population dynamics within each component.

The first step in developing MICE is to articulate the problem being addressed. Unlike some whole-of-system models, MICE are not designed to have general applicability, and the selection of model components (e.g. the number of biological components and spatio-temporal strata) should reflect the problem being addressed. The data used to parameterize MICE (Fig. 2) include many of the same data sources (catches, catch rates, catch and survey age and size composition) used when fitting single-species models, but also include sources detailing relationships between environmental data and biological processes as well as predation. Data represented by the overlap of categories can provide information on key processes in the interaction of model elements (e.g. data on quota trading represent a nexus between social networks and market forces; data on temperature-dependent growth rates (i.e. T°-Growth) represent a nexus between physical environmental parameters and the basic biology of resource species) (Fig. 2). The values for the parameters are then estimated by maximizing the joint likelihood of the parameters given the data. An underlying principle is to use simpler representations to reduce model complexity while still having sufficient structure and detail to achieve good fits to the available data (Table 2).

Sharks, myctophids and squid are all treated as generic species groups in the Coral Sea model, adhering to the general principle of not aggregating serially linked groups (predator and prey), or species, age-classes or functional groups with rate constants which differ by more than twofold to threefold (Fulton *et al.* 2003). The system contains species of high commercial value (tuna) as well as species of conservation concern

such as the blue shark (*Prionace glauca*, Carcharhinidae). The species modelled are yellowfin tuna (*Thunnus albacares*, Scombridae) and bigeye tuna (*Thunnus obesus*, Scombridae), and the billfish species are swordfish (*Xiphias gladius*, Xiphidae) and striped marlin (*Tetrapturus audax*, Istiophoridae) Species-specific splits are used in these cases because there are sufficient data to do so, and trying to account for obviously different abundance trends, fishery selectivities and life-histories in a single model component would be difficult. This structure was also chosen based on the rule of thumb that at least 90% of the mortality (due to fishing and predation) on each of the key species groups, including the key prey species (squid and myctophids), is captured by the model. Reducing the number of modelled species reduces the number of inter-species links that need to be modelled. However, reducing the number of components and linkages may lead to difficulties interpreting key observational data and incorrect inferences regarding the behaviour of the system (Yodzis 1998).

Exports of freshwater flow from catchments are used as the key driver and integrator of the catchment-to-coast-marine ecosystem in the GoC model. The fishery in the GoC targets several species of prawns, with the major species being common banana prawns (*Penaeus merguiensis*, Penaeidae), grooved tiger prawns (*Penaeus semisulcatus*), Penaeidae) and brown tiger prawns (*Penaeus esculentus*, Penaeidae). The blue endeavour prawn (*Metapenaeus endeavouri*, Penaeidae) is a key by-catch species. In addition to prawns, the GoC model also explicitly represents key predators and food sources for these prawn species (with explicit modelling of microphytobenthos and meiofauna) (Fig. 1b) (Burford *et al.* 2009). The model was constructed by starting from a few species and processes, and then gradually adding species and complexity to adequately explain the system dynamics.

Modelling each component

Once the key species have been identified, it is then necessary to decide how each will be modelled. In general, more complexity will lead to greater challenges in parameterizing the model, quantifying uncertainty and evaluating fit diagnostics. Consequently, in MICE, some populations are modelled by age, length and stage, and others at a more coarse resolution. In general, there will be more structure for populations for which more data are available. The simplest case is that of no structure at all, where a population is modelled in totality (either in terms of numbers or biomass). However, most real populations exhibit structure in terms of life-stage, age or length. If an ecological process is governed by age/length, then it is logical to use an age- /length-structured framework. However, an age-only or length-only model would be appropriate when age and length are interchangeable or when it is impossible to determine age. In relation to biodiversity management, there is an increasing body of work that identifies the advantages of being able to estimate the community-level length distribution, and ecosystem indicators that can be derived from such a distribution (Blanchard *et al.* 2010, 2011). Tunas and billfishes are represented using age-structured equations in the Coral Sea model, while sharks are modelled using a two stage (mature and immature) model. Squid and myctophids are modelled using simple aggregated delay-difference models because of lack of information.

At its core, the GoC model is based on the prawn size-structured population dynamics model of Punt *et al.* (2010), although *M. endeavouri* is replaced by *P. merguiensis*. Fish predators, barramundi (*Lates calcarifer*, Latidae) and threadfin salmon (*Polydactylus macrochir*, Polynemidae), were added because these are the only estuarine species with long-term species-specific catch data. Other species such as sharks and rays that are known to be important prawn predators are either not targeted or ignored as there are no species-specific catch data (Fig. 1b). A Schaefer production model was used for threadfin salmon and an age-structured model for barramundi. The prawns' food sources (macrophytobenthos and meiofauna) were modelled using discrete growth equations with fitted functional forms relating growth rate to a range of physical parameters, in particular salinity and nutrient concentration.

Temporal and spatial resolution

In general, the finer the temporal resolution, the more accurately the dynamics can be represented. However, this comes at the cost of an increased computational burden and the need for more detailed data. The time steps for each of the com-

ponents may differ. For example, short-lived target species – such as prawns – might need to be modelled with weekly time steps, while other groups such as sharks, whose population sizes do not change as rapidly, could be modelled on an annual time step. Spatial structure is seldom accounted for in traditional single-species models. However, accurately modelling the dynamics of trophic interactions and fishing activity may require spatial structure.

The time step in the Coral Sea model is quarterly to match the availability of data, and the model includes several spatial strata. The GoC adopts the same weekly time step for prawns as in Punt *et al.* (2010), but uses an annual time step for the fish given this is the scale at which data are available for these groups. The model is spatially structured to represent two stocks of banana prawns and to capture differences in the functioning of the more northerly (Weipa) and southern (Karumba) stocks. Physical variables such as flow volume and salinity are averaged over each of the northern and southern regions, which consequently each have separate inputs.

Modelling fish movement

Including spatial structure in traditional stock assessments is recognized as important (Stefánsson and Palsson 1998; Begley 2005), and some stock assessments include multiple spatial strata, with movement of animals among them (e.g. Fournier *et al.* 1998). While stock assessment packages such as stock synthesis (Methot 2005, 2009) and CASAL (Bull *et al.* 2004) include the ability to model spatially structured populations, doing so is still the exception rather than rule primarily because of a lack of data to estimate movement rates. Tag-recapture data have traditionally been most informative regarding movement, but there is no guarantee that one can adequately estimate movement parameters even when tagging data are integrated with catch, effort and even survey information. More recently, methods have been developed to utilize data from electronic tagging technology in population models with movement dynamics (e.g., Eveson *et al.* 2012). However, there are still major challenges to be overcome even in data-rich scenarios. While MICE may lack the spatial resolution of whole-of-ecosystem models, they are still better placed than most stock assessment approaches to deal with spatial population processes.

Models of animal movement behaviour can be somewhat simplistically classified into two main categories. Time-invariant (or random) movement is driven primarily by diffusive processes whereby a fixed proportion of animals move from one area to another over a given timescale (which can be asymmetric to account for spatially and life-history-related anisotropic effects). Alternatively, dynamic movement involves time-dependent factors, such as reproduction or prey/competitor density and environmental conditions (Hobday and Hartmann 2006), impacting rates of movement. Dynamic movement can be modelled using preference or habitat suitability functions whereby each spatial region is assigned a level of preference (conditional on the key movement-influencing factors) for a particular animal of a given species/age/length/area etc. and a probabilistic movement model constructed based on the relative preference of each area by that animal (Bentley *et al.* 2004).

Both types of movement are explored in the Coral Sea model. In contrast, movement is not represented explicitly in the GoC; rather, simple rules are used to capture important processes – for example, if freshwater flows exceed a threshold value, juvenile prawns are assumed to move or be flushed out to the offshore area, and hence be subject to different dynamics.

Species interactions

One of the greatest challenges in moving from single-species to ecosystem models is the choice of the functional form(s) to represent inter- and intra-species trophic (and to a lesser extent non-trophic) interactions (Plagányi 2007). Predation, cannibalism and competition are the three kinds of species interactions that warrant consideration in MICE. Predation, including cannibalism, is the fundamental interaction that needs to be represented when moving beyond single-species models. Past multispecies methods (e.g. MSVPA, Magnússon 1995; Jurado-Molina and Livingston 2002; MSM Jurado-Molina *et al.* 2005; MALMAL Kinzey and Punt 2009) have assumed a one-way interaction between predator and prey, that is, reduced numbers of prey consumed do not functionally impact the predator growth rate. In contrast, models based on the principles of conservation of matter include two-way feedback by assuming that prey eaten is converted into

predator biomass or numbers. The functional form used to represent such feedbacks is both critical to the system dynamics and difficult to estimate (Hunsicker *et al.* 2011). However, MICE can avoid this by considering what consumption affects (i.e. survival, reproduction and growth rate) rather than consumption itself. Plagányi and Butterworth (2012) represent the dependence of predators on prey using a survival and reproductive success factor that is a function of the available biomass of prey and acts as a multiplier to the juvenile recruitment parameters, namely the reproduction and/or the juvenile survival rate. Thus, the predator-prey interaction term links predator breeding success directly with prey abundance, avoiding the need for an explicit consumption-related term. A single parameter controls the level of prey abundance (relative to the single- or multispecies carrying capacity) below which there is an appreciable negative impact on predator breeding success.

In a MICE context, it is key to account for the uncertainty in the nature and form of the interactions modelled, as well as the two-way feedback between predators and prey. Ideally, the parameters and form of the predator-prey functional response should be estimated by fully integrating predation and diet data in the parameter estimation process (e.g. Kinzey and Punt 2009). If no suitable data are available, an alternative would be to assess the robustness of model outputs to a range of alternative functional response formulations (see Kinzey and Punt 2009; Koen-Alonso and Yodzis 2005; Hunsicker *et al.* 2011; for an overview of functional response types, the current state of the art and the challenges in their use). A third approach would be to bound the uncertainty in the predator-prey relationship; Plagányi and Butterworth (2012), for example, propose the use of a generalized functional response curve that is parameterized to bound the uncertainty of the likely level of sensitivity of a predator to its prey.

There are also consequences of competitive effects on growth and survival rates. For example, an increase in growth rate because of decreased competition can result in an increase in effective survival rates-at-age as animals grow faster out of the key predator-vulnerable length classes. From a more predator-focussed viewpoint, decreased competitor abundance could result in increases in survival rates if mortality due to prey scarcity is important, but one might expect growth rates to increase if prey search and process time (and energy expended) decreases with competition. The end result from certain viewpoints is the same, such as an increase in the probability of future biomass growth of the benefitting competitor, but the functional inclusion of each process in a MICE, and their potential impact on the whole model's dynamics, could be quite different.

Representing predation by tuna in the Coral Sea model first involves computing the relative preference for each prey species. The overall prey availability, as a proportion of a combined prey carrying capacity, is then computed as the sum of the products of the individual preferences and prey biomass. A normalized sigmoid function, with output range 0–1, and depending on overall prey depletion, is then used to represent the dependence of reproduction on prey abundance. The parameters of this relationship could ideally be estimated by fitting to data. As there are fewer data for the sharks, a single threshold prey abundance below which reproductive potential decreases linearly to zero for zero prey abundance is used, meaning that only one single parameter is needed to describe the relationship.

For the GoC model, it appears that the growth and survival of barramundi and threadfin salmon may be higher when prawns are plentiful, but that these predator species may not have much impact on prawn abundance (Robins *et al.* 2006). The GoC model thus considers a one-way interaction term only. However, fish growth rates are also modelled as dependent on flow, based on formulations available in the literature (Balston 2009).

Modelling recruitment

The starting point for modelling recruitment in MICE is to draw on the methods as used for single-species models. For example, formulations such as the Beverton–Holt or Ricker stock-recruitment relationships, with associated stock-recruitment deviations can be fitted. The same principles as for single-species models also apply with respect to handling of uncertainty about recruitment, namely that the sensitivity of model predictions to alternative plausible recruitment hypotheses needs to be demonstrated.

There is added complexity when modelling recruitment within a multispecies system or one with multiple spatial strata. For example, one of the parameters of a conventional stock-recruit-

ment relation is carrying capacity. However, in a multispecies system, the carrying capacity for one species may be negatively impacted by increased abundance of another species that shares an ecologically similar niche. Similarly, deviations in recruitment about the stock-recruitment relationship may be correlated among species owing to such deviations being driven by common environmental factors (Cury *et al.* 2008). Allowing for correlation in recruitment deviations among species has the added advantage that information from more data-rich species can inform recruitment for more data-poor species. Conventional stock–recruitment relationships can be generalized, as outlined above, to allow the reproductive or survival rate to depend on consumption of prey. Finally, spatial variation in recruitment success or the transport of recruits to other regions could be important in some instances. In the MICE framework, this would involve fairly simple relationships or rules to modify spatial recruitment patterns, rather than detailed modelling of, for example, physical transport processes.

A single stock-recruitment relationship is implicitly assumed for multiple species in the Coral Sea model because species are modelled as aggregated or functional groups (sharks, squid and myctophids), so that groups where such a relationship is modelled explicitly will share common parameters. Recruitment of banana prawns in the GoC model is determined by a stochastic stock-recruitment relationship where the deviations from deterministic expectations are based on relationships to physical (flow volume) and biological (availability of meiofauna) components.

Representing human behaviour

Human behaviour has received the least attention in applied fisheries assessment models, despite growing recognition of its importance for both tactical and strategic management advice (Wilen *et al.* 2002; Fulton *et al.* 2011). Human decision-making determines the spatial and temporal distribution of fishing mortality on a variety of scales, from hourly decisions related to the operation of fishing gear, to daily or weekly decisions on the duration and location of fishing trips, to the seasonal choices of fishing activities and to the longer-term decisions associated with entering or exiting a fishery, and investing into fishing vessels (see Nostbakken *et al.* 2011; van Putten *et al.* 2011 for recent reviews).

Although the explicit inclusion of human behaviour in assessment models remains the exception, MICE can rely on research on the empirical modelling of fleet dynamics using, for example, location-choice models (e.g., Smith and Wilen 2004; Vermard *et al.* 2008), to capture such processes. Consistent with the MICE approach, bio-economic models that include human behavioural components have usually been based on simpler representations of biological systems, as has been the case in the broader domain of biodiversity conservation (Eppink and van den Bergh 2007). For example, within Australia, efforts to include a representation of fleet dynamics in bio-economic fisheries models have been based on single-species assessments (Hamon *et al.* 2010), spatial meta-population models (Little *et al.* 2009) and models of multiple species linked through technical interactions (Punt *et al.* 2011a). Other approaches have considered the explicit modelling of demand for fish products and its impacts on prices driving harvesting patterns, using simplified representations of interconnected fisheries markets at the international level (Mullon *et al.* 2009).

In the Coral Sea model, the fishery is decomposed into fleets (with grouping based on similar gear characteristics and targeting practices) for consistency with the stock assessments for the tuna fisheries in the Coral Sea. Location, species preference and economic information form the basis for fleet behaviour in future projections, along with scenarios regarding management measures.

Data and parameter estimation

The values for some of the parameters of a model (e.g. growth rates) will be pre-specified based on auxiliary information (e.g. analyses of data sets specific to those parameters or using heuristics and inferences for other systems). However, the bulk of the parameters should be estimated by maximizing the likelihood function. Likelihood functions have been derived from hypothesized or assumed sampling distributions for many of the types of data that would be included in MICE (see Francis (2011) for a recent summary). Depending on the way they are summarized, diet data can be included in the likelihood function analogously to

how fishery catch data are treated (Kinzey and Punt 2009). It is common to include prior information (e.g. on growth and predation rates, assimilation efficiencies) in addition to data for the system under consideration in the objective function to be optimized. When multiple data sources are available, each needs to be weighted. In cases in which the data are somewhat contradictory, this needs to be done carefully (Francis 2011), and sensitivity may need to be explored to different data weighting schemes.

The data used when fitting MICE will include those on which single-species assessments are typically based (see Fig. 2). MICE make use of catch data, at least one index of relative abundance (either from catch rates or from surveys), and compositional data for the catch (and any surveys). Catch length-frequency data are available for most stocks, but generally catch age-frequency data are more informative regarding the dynamics of populations.

Environmental data are rarely included in single-species stock assessments (the assessment of sablefish (*Anoplopoma fimbria*, Anoplopomatidae) off the U.S. west coast (Schirripa 2007) being a noteworthy exception), although frameworks to achieve this have been developed (e.g., Maunder and Watters 2003; Haltuch and Punt 2011; Maunder and Deriso 2011) and could be used when fitting MICE in which there are environmental drivers of biological processes. Average sea surface temperature has been used as a proxy for the fishing mortality at which maximum sustainable yield (MSY) is achieved (F_{msy}) in the harvest control rule for Pacific sardine (*Sardinops sagax*, Clupeidae) (Hill *et al.* 2008) and this could be accomplished for MICE that integrate environmental drivers.

Data on predation are not explicitly included in most single-species stock assessments. Instead, if predation data are used in such assessments, it is by taking the estimates of natural mortality from multispecies models as fixed values in the single-species assessment (e.g. ASMFC 2010). However, data from diet studies can be summarized in ways similar to those from fisheries by treating each predator age/size class as a 'fleet' [e.g. using estimates of consumption of each prey species by each predator age/size class, and data on the proportion of each prey age/size class in the diets of each predator age/size class when fitting models (Kinzey and Punt 2009)].

Calibration of the fishing behaviour components of MICE will usually rely on similar data to those used in single-species assessments. For short-term effort allocation modelling, the data will usually include fleet/vessel technical characteristics, catch rates and, where relevant, output prices per category of landed catch, as well as cost differences across fishing options. Modelling of other short-term behaviours such as discarding, compliance with fishing regulations or leasing of catch rights may require additional economic information, for example, regarding the penalties (financial or other) associated with alternative behaviours or the lease prices of catch rights. If available, cost and earnings data related to the fishing fleets can be used to provide an assessment of the economic outcomes associated with alternative states of the ecosystem and the fishery.

A concern is that data, usually collected for particular species, will often be aggregated into functional groups in MICE. The utility of catch-rate indices and even catch and survey age and length frequencies is reduced because it is no longer clear how to compute the model predictions associated with the available data. The relative size of each of the populations, as well as any type of differential targeting behaviour, will be crucial in interpreting aggregated catch-rate data. Moreover, aggregating data over multiple species can lead to parameters such as the catchability for a functional group varying over time even if the value of the parameter concerned is time-invariant for each of the species in the group, owing to changes over time in the relative abundance of the species in the group. Alternatively, one species in a functional group could be modelled separately, and parameter values mirrored for all species. This treatment would lead to a similar number of estimable parameters while modelling changes in species composition over time. The Coral Sea model explores these challenges, given that commercial catch-per-unit-effort data are the most abundant and (usually) influential source of data on trends for the tuna, sharks and billfish.

The parameters of the GoC model are estimated using catch, catch rate and length-frequency data from surveys and the fishery, and tag release-recapture data (Punt *et al.* 2010). Available catchment data for the regions in the model are used as input variables and drivers. Modelled changes in monthly microphytobenthos and meiofaunal biomass in normal and flood years are compared with

Table 3 A subset of existing minimally realistic models that could form the basis for tactical management advice.

No	Reference	Region	Species modelled
1	Kinzey and Punt (2009)	North Pacific	Walleye pollock, Pacific cod, Atka mackerel (*Pleurogrammus monopterygius*, Hexagrammidae)
2	Hollowed *et al.* (2000)	North Pacific	Walleye pollock, arrowtooth flounder (*Atheresthes stomias*, Pleuronectidae), Pacific halibut (*Hippoglossus stenolepsis*, Pleuronectidae), Steller sea lion (*Eumetopias jubatus*, Otariidae)
3	Tjelmeland and Lindstrøm (2005)	NE Atlantic	Minke whales (*Balaenoptera acutorostrata*, Balaenopteridae), herring (*Clupea harengus*, Clupeidae)
4	Livingston and Jurado-Molina (2000)	North Pacific	Walleye pollock, Pacific cod, Greenland turbot(*Reinhardtius hippoglossoides*, Pleuronectidae), yellowfin sole (*Limanda aspera*, Pleuronectidae), arrowtooth flounder, northern fur seal (*Callorhinus ursinus*, Otariidae)
5	Lindstrøm *et al.* (2009)	NE Atlantic	Herring, capelin, Atlantic cod, minke whales
6	Lehodey *et al.* (2008)	Pacific	Skipjack tuna (*Katsuwonus pelamis*, Scombridae), yellowfin tuna, albacore tuna (*Thunnus alalunga,* Scombridae)
7	Jurado-Molina *et al.* (2005)	North Pacific	Walleye pollock, Pacific cod
8	Garrison *et al.* (2010)	NW Atlantic	Atlantic menhaden (*Brevoortia tyrannus*, Clupeidae), striped bass (*Morone saxatilis*, Moronidae), weakfish (*Cynoscion regalis*, Sciaenidae), bluefish (*Pomatomus saltatrix*, Pomatomidae)
9	Mohn and Bowen (1996)	NW Atlantic	Atlantic cod, grey seals (*Halichoerus grypus*, Phocidae)
10	Mori and Butterworth (2006)	Antarctic	Blue whale (*Balaenoptera musculus*, Balaenopteridae), minke whale, humpback whale (*Megaptera novaeangliae*, Balaenopteridae), Antarctic fur seal (*Arctocephalus gazelle*, Otariidae), crabeater seal (*Lobodon carcinophagus*, Phocidae), krill
11	Blamey (2010); Blamey *et al.* (in press)	South Atlantic	Abalone, Cape rock lobster, urchin
12	Danielsson *et al.* (1997); Hvingel and Kingsley (2006)	North Atlantic	Shrimp, cod

observed values (Melissa Duggan, personal communication).

Existing MRMs and their relationship to MICE

In the following section, we identify some general properties of the current state of MRMs, and how MRMs relate to the MICE concept outlined above. As noted in the introduction, MICE are similar to MRMs. While MRMs are capable of producing outputs that could be used for tactical decision-making, this appears to be a rare occurrence – noting that only one of the MRMs (#12 in Table 3) has been used to provide tactical management advice (NAFO 2006).

Several existing MRMs have been fit to available data and superficially seem to be capable of providing estimates of biomass, reference points, etc. (Table 3). However, most existing MRMs have been developed as the basis for strategic evaluations (e.g. Punt and Butterworth 1995; Schweder *et al.* 2000; DeYoung *et al.* 2004; Watters *et al.* 2005, 2006; Plagányi and Butterworth 2008; A'mar *et al.* 2009; Howell and Bogstad 2010). Some of these MRMs (e.g. Punt and Butterworth 1995; Plagányi and Butterworth 2008, 2012) could have been used for tactical advice (catch limits for the South African fur seals (*Arctocephalus pusillus*, Otariidae); approaches to spatially allocating catches of krill (*Euphausia superba*, Euphausiidae) in the Southern Ocean), but have been omitted from Table 3. In addition, other published MRMs (e.g., Bogstad *et al.* 1997; Koen-Alonso and Yodzis 2005) are primarily simulation models with parameters set based on auxiliary information. In principle, the values for the parameters of these MRMs could have been estimated rather than being pre-specified. Finally, whole-of-ecosystem model software (such as EwE) could be used to implement MRMs and hence act as MICE. However, these models have rarely been used this way outside of training exercises.

Eight of the 12 MRMs in Table 3 are variants of multispecies virtual population analysis (MSVPA)

(or 'integrated' analogues), while the other four (#6, #10, #11, #12) are more custom-designed models. MSVPA uses commercial fisheries catch-at-age and fish stomach-content data to estimate both past fishing mortalities and the predation mortalities on some of the chief fish species of interest (Sparre 1991; Magnússon 1995). The approach splits natural mortality into two components: predation owing to predators explicitly included in the model (*M2*), which depends on time and age because of variations in predator abundance, and residual mortality (*M1*) owing to all additional factors that are customarily taken to be constant. Perhaps not unexpectedly given the availability of data worldwide, the bulk of the models are for the North Atlantic and North Pacific, with a focus on species of major commercial importance [cod (*Gadus morhua*, Gadidae) in the North Atlantic and walleye pollock (*Theragra chalcogramma*, Gadidae) in the North Pacific]. The number of species modelled range from two to six among the 12 MRMs, well below the number typically considered in whole-of-ecosystem models. Six of the 12 MRMs include a marine mammal, highlighting that many of these models were developed to address questions related to the impact of marine mammals on fishery yields.

Although all 12 MRMs were fitted to the types of data used in typical stock assessments (catches, abundance indices, catch age/length compositions), only two of the MRMs (Hvingel and Kingsley 2006; Kinzey and Punt 2009) were fitted to diet data. The studies based on MSVPA used diet data (in particular assumed values for rations), but were not formally fitted to these data (they did not fit to catch age composition data either, assuming instead that these data are measured exactly). Although the GADGET framework (http://www.hafro.is/gadget, accessed 27 April 2012) on which Lindstrøm *et al.* (2009) was based can represent spatial structure, only one of the models (Lehodey *et al.* 2008) was spatially structured. However, this MRM was not fitted to movement data. A study by Lehodey *et al.* (2008) was also the only MRM that considered habitat impacts and environmental drivers, although most of the MSVPA-type models estimated annual recruitment by species so implicitly accounted for environmental impacts on recruitment. None of the MRMs provided output in the form of economic indices, highlighting the general lack of linkages between MRMs and economic models.

From principle to practice in tactical applications of MICE

Tactical applications of ecosystem models are extremely rare, and such applications will likely always be the exception rather than the rule because it isn't necessary or ideal to base tactical management advice on the outputs of ecosystem models in most instances. Here, we provide examples of tactical advice that has been applied, followed by some examples of tactical applications that we consider achievable in principle. Well-constructed MICE are based on extensive stakeholder consultation (evident too from the question-driven and restricted focus principles), and this is critical to gaining their acceptance as tactical management tools.

The estimation of current abundance and exploitation rate is central to providing tactical management advice, and multispecies models have been used to compare and assess the accuracy of quantities derived using single-species models. Attempts to refine time-invariant natural mortality estimates in single-species assessment models, with updates tuned to temporal variations in predators, remain one of the few tactical applications where the outputs of a multispecies model has contributed directly to the assessment of the status and productivity of a resource (e.g. ASMFC 2010). Multispecies statistical models such as those of Livingston and Jurado-Molina (2000) and Jurado-Molina *et al.* (2005) were developed with the objective of examining trends in mortality due to predation, but these applications are largely strategic rather than tactical to date. A rare tactical example is provided by the use of Gadget (and other frameworks) to model the cod-shrimp interactions as a part of the Icelandic cod assessment process in some years. For example, Danielsson *et al.* (1997) used a bio-economic model of cod, capelin (*Mallotus villosus*, Osmeridae) and shrimp (*Pandalus borealis*, Pandalidae) to support the choice of a lower total allowable catch (TAC) for cod to allow cod to recover, while highlighting the predicted resultant large reductions in capelin and shrimp catch.

There are a growing number of countries in which tactical management advice depends on comparisons with biological reference points, such as the spawning biomass level at which MSY is achieved. Several recent studies have attempted to provide guidance on these reference levels based on consideration of trophic interactions and

dependencies (Smith *et al.* 2011; Tyrrell *et al.* 2011; Pikitch *et al.* 2012). MICE are appropriate tools for contributing to understanding the choice of reference points based on ecosystem considerations. The optimal values of reference points can be expected to vary in synchrony with variability in an ecosystem (Collie and Gislason 2001), and hence there is potential for more tactical applications of MICE to attempt updated estimation of reference points.

Considerable research has focused on attempts to derive predictive relationships between environmental variables and recruitment success, but there is still a paucity of examples that have successfully integrated these into tactical management advice (Schirripa 2007 being the noteworthy exception). Habitat prediction models such as those combining data from an ocean model and pop-up satellite archival tags (Hobday and Hartmann 2006; Lehodey *et al.* 2008) can provide support to the management of species such as southern bluefin tuna (*Thunnus maccoyii*, Scombridae), but these approaches have not yet been implemented in MICE. Finally, MICE can be used to estimate similarities in recruitment deviations among species within a common region, and such estimates could subsequently be used within a 'Robin Hood' approach to fisheries (Punt *et al.* 2011b; Thorson 2011) to inform recruitment for information-poor species.

Spatial management is a fourth important tactical management advice category where MICE are playing an increasing role. An example is the use of ecosystem models (of intermediate complexity) by the Commission for the Conservation of Antarctic Marine Living Resources to inform the spatial allocation of krill catches to reduce the potential impact of krill fishing on dependent predators in the Antarctic system (Plagányi and Butterworth 2008, 2012; Watters *et al.* 2008). Actual implementation of a spatial subdivision rule informed by multispecies models is still some way off, but consideration is being given to moving to a dynamic allocation based on land-based predator monitoring conducted just prior to or early during the fishing season (Hewitt *et al.* 2004). Also, no-take marine protected areas are being implemented widely, although the current generation of single-species assessment models generally ignores spatial closures. An exception to this is the assessment for southern rock lobsters (*Jasus edwardsii*, Palinuridae) off Victoria, Australia, that models the components of the population in the fished and unfished areas explicitly (e.g. Hobday *et al.* 2005). The Coral Sea model has been modified to include an evaluation of a marine protected area.

Efforts or mandates to rebuild fish stocks are an important component of many fishery management systems. For example, the U.S. Magnuson-Stevens Fishery Conservation and Management Act (U.S. Public Law 104-297) mandates ending overfishing and rebuilding depleted fishery resources. Attempts to rebuild stocks can be complicated by confounding between natural and anthropogenic sources of mortality, such that MICE could potentially be used to inform evaluation of rebuilding strategies. MICE that have been used to understand the underlying processes mediating stock decreases, as well as the timescale and likelihood of future recovery, include that applied to the South African abalone (*Haliotis midae*, Haliotidae), Cape rock lobster (*Jasus lalandii*, Palinuridae) and urchin (*Parechinus angulosus*, Parechinidae) regime shift off South Africa (Blamey 2010; Blamey *et al.* in press).

To the best of our knowledge, tactical applications that include either human behavioural components or an economic assessment of the outcomes of management scenarios have yet to be developed. Bio-economic modelling approaches exist on the basis of which these applications could be developed. For example, the analysis of feasible rebuilding strategies may need to account for the possible constraints and inertia in adjustments of fishing capacity because of economic, social and political reasons. The trade-offs between commercial and non-commercial values associated with harvesting of a marine ecosystem can also be accounted for (Martinet and Blanchard 2009).

Discussion

Advantages and disadvantages of MICE

Models of Intermediate Complexity for Ecosystem assessments suffer from many of the same disadvantages as other modelling approaches, but have a number of key advantages in the context of providing ecosystem management advice. This is largely because they can provide tactical and not just strategic advice, focus more specifically on a restricted subset of an ecosystem and account more fully for uncertainty than whole-of-ecosystem models (Table 1). MICE can include rigorous

methods to address the high levels of uncertainty associated with ecosystems (with regard to their structure, input data and model parameters). The use of standard data fitting methods, familiar to fisheries scientists and managers, is another advantage. Although the approach is highly data intensive, an advantage is that the models can be structured to take account of the available information, as illustrated by the Coral Sea and GoC examples. The aim of including these examples is not to draw conclusions or provide estimates of management-related quantities, but rather to illustrate the model-building process and identify the key questions that need to be asked when constructing MICE.

Models of Intermediate Complexity for Ecosystem assessments lack the level of detail, system components and realism required for evaluating complex systems. However, the closer focus of MICE on a subset of the ecosystem is advantageous in terms of the potential both to contribute to tactical decision-making and to improve understanding of components of the ecosystem. Potential biases that arise through restricting the focus and scope of MICE can be evaluated by comparing results from intermediate and highly complex models of the same system. The weakest point of MICE is the possibility of choosing the wrong key species and interactions. Ways to reduce this risk include the availability of good data and information for a system, broad consultation with stakeholders, cross-comparisons with alternative MICE (for example, models with a different underlying structure), as well as with single-species and ecosystem models of the same system (Plagányi 2007; Fulton 2010), testing the robustness of model results across a broad range of uncertainties and maintaining a healthy scepticism about one's own models.

There is considerable value (e.g. cross-comparisons, error checking, speed and ease of use, resources to develop and refine an approach) in using the same platform to develop a broad range of models, as has been achieved for example with EwE (Walters *et al.* 1997). By comparison, MICE are usually labour-intensive as individual applications are typically custom-built, and hence a more in-depth analysis of these models, agreement on the underlying philosophy and some generic rules for their development will assist in ameliorating the challenges.

The flexible nature of MICE is advantageous in tailoring models to specific systems, problems and species. There are no restrictions because of using modelling packages that lack transparency or the ability for a user to understand and make changes to the behind-the-scenes formulation of a model, or due to generic formulations that are not suited for a particular species or situation. MICE do, however, require redesign and reparameterization if the nature of the question changes. By comparison, end-to-end modelling allows for a much wider range of response variables, functional groups, *et cetera* to be investigated (Rose *et al.* 2010; Fulton 2010).

Stakeholder consultation

Stakeholder participation and dialogue are integral components of an EAF and are similarly important throughout the conception, development and implementation of MICE. MICE have an advantage over more complex ecosystem modelling approaches in that they are conceptually more tractable for stakeholders and focus more clearly on specific problems identified by stakeholders. The advantages to engaging stakeholders range from broadening the underlying level of conceptual understanding of the system to achieving stakeholder buy in, resulting in greater likelihood of uptake of results arising from the modelling. A key advantage is speed of computation and modification of models, making it ideal for a participatory setting with stakeholders.

Integrated fisheries and conservation models for management

An important emerging issue is the need to integrate fisheries and marine conservation issues in science-based decision-making (Salomon *et al.* 2011). MICE-type models have already been used for this purpose because many of them have been constructed to address questions such as the competition between marine mammals and fisheries (Table 3), and hence include both fish and species of conservation concern. Models such as GADGET, Bormicon (Bogstad *et al.* 1997), MULTSPEC (Bogstad *et al.* 1997), Scenario Barents Sea (Schweder *et al.* 2000), Seastar (IWC 2004) and FLEXIBEST (IWC 2004) are all MICE-like, have variously incorporated predation of fished species and their predation by marine mammals, and are all spatially structured given the importance of considering spatial-temporal overlaps between fish-

eries, marine mammals and shared prey species. MICE have considerable potential to bridge the divide between fisheries and conservation managers because they are capable of simultaneously assessing the sustainability of a fishery, the major ecosystem impacts of fishing, and need to meet conservation goals in a timely way (both in terms of computation and the length of development/implementation). MICE can better inform management of marine systems because they apply similar levels of rigour to assessments of conservation species as is the norm in single-species assessments of fish species.

Including social and economic dimensions

Given the importance of social and economic dimensions in ecosystem-based approaches to fisheries management (National Research Council 2006), developing these dimensions in MICE is bound to strengthen their usefulness. Further development can rely on conceptual approaches that are already well established, including the bio-economics of harvesting predator-prey systems (Hannesson 1983; Flaaten 1998; Sanchirico and Wilen 2001; Boncoeur *et al.* 2002), or the inclusion of habitat impacts of fishing in bio-economic models (Imeson and van den Bergh 2004). To date, however, applications to specific empirical contexts remain limited. This is probably due to a lack of adequate empirical information on the social and economic processes that the models would need to represent. It may also be because including economic and social decision processes in ecosystem models can drastically increase model complexity. By focusing on specific decision processes that are seen to determine the outcomes of interest, MICE reduce this complexity and may facilitate the inclusion of human dimensions in ecosystem models. In addition, they can expand how the results of MICE are reported, to include social and economic indicators.

Of MICE and MSE

Models of Intermediate Complexity for Ecosystem assessments can and have been used as operating models as part of MSE. These frameworks involve simulation testing of the implications, for both the resource and its stakeholders, of alternative combinations of monitoring data, analytical procedures and decision rules. MSE (Butterworth and Punt 1999; Smith *et al.* 1999; Sainsbury *et al.* 2000; Rademeyer *et al.* 2007) is being increasingly used to evaluate strategies for managing ecosystems. One strength of MSE is that it is consultative – all stakeholders can have input into the candidate operating models and management scenarios. As the approach demands clear objectives to judge the performance of the alternative strategies, the method forces participants to be clear about their objectives and to specify relevant performance indicators. Significant challenges remain, however, in determining and quantifying appropriate management objectives that take into account diverse societal goals or ecosystem metrics (Murawski 2000; Marasco *et al.* 2007).

There is a natural relationship between MICE and MSE because both approaches have a formal treatment of uncertainty at their core, consistent with the Precautionary Approach (FAO 1996; Butterworth 2007). Key features of these applications are that they rely on a feedback control mechanism to adjust for inevitable errors in current perceptions about a system and have the ability to explore management constraints and issues of uncertainty (Walters 1986; Cooke 1999; Butterworth 2007). As an example, Punt and Butterworth (1995) explicitly accounted for uncertainty and management issues through the use of a simulation framework that incorporated the feedback control rules actually in place for setting the TAC for a hake (*Merluccius capensis*, Merlucciidae and *Merluccius paradoxus*, Melucciidae) fishery. Thus, they evaluated whether the management system applied to compute TACs was able to take advantage of any increases in hake sustainable yields caused by changes in South African fur seal numbers. The use of MICE as operating models in MSE testing is a powerful tool for taking into account a broad range of uncertainties, at all levels from parameter uncertainty, model structural uncertainty to implementation uncertainty. This renders it a robust method that is more likely to be accepted by most stakeholders and has the potential to significantly advance ecosystem-based fisheries management.

Most MSE studies are based on management strategies in which the assessment method is a single-species approach. However, MICE could be included in MSE studies as assessment models, thereby allowing the benefits or otherwise of using MICE for tactical decision-making to be evaluated.

Into the future with MICE

Models of Intermediate Complexity for Ecosystem assessments provide the ability to address both strategic and tactical questions, although there are few examples of the latter [Hvingel and Kingsley (2006) and ASMFC (2010) being noteworthy exceptions]. Moving forward, we see an increasing role for MICE in addressing tactical issues, such as estimation of current abundance, exploitation rates and reference points for a connected suite of species in an ecosystem. Other important applications include the prediction of future recruitment (and hence sustainable catches/potential rebuilding rates) given environmental factors and the implications for yield and population structure arising from closed areas. We acknowledge that, as with most other ecosystem models, the focus of MICE to date has been largely on the biophysical and ecological compartments and processes, with human dimensions absent, or only sparsely considered. As theories and models of resource use dynamics are advanced, we anticipate and recommend that MICE more fully embrace economic and ecological-economic modelling principles (Fig. 1). Given the question-driven focus of MICE, it may be more appropriate to adopt the social and economic considerations that drive management as a starting point for the development of some models.

In summary, we have shown that a group of models, which we refer to as MICE, 'Models of Intermediate Complexity for Ecosystem assessments', attempt to estimate the parameters of multispecies models by fitting to data, to apply statistical diagnostic tools to evaluate model performance, and to account for a broad range of uncertainty. These models are beginning to be applied for tactical as well as strategic fisheries decision-making that takes into account the broader ecosystem considerations given a few well-defined objectives.

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References

A'mar, Z.T., Punt, A.E. and Dorn, M.W. (2009) The evaluation of two management strategies for the Gulf of Alaska walleye pollock fishery under climate change. *ICES Journal of Marine Science* **66**, 1614–1632.

ASMFC (2010) Stock Assessment Report No. 10-02 of the Atlantic States Marine Fisheries Commission Atlantic Menhaden Stock Assessment and Review Panel Reports. 1–325.

Balston, J. (2009) Short-term climate variability and the commercial barramundi (*Lates calcarifer*) fishery of north-east Queensland, Australia. *Marine and Freshwater Research* **60**, 912–923.

Begley, J. (2005) *Gadget User Guide*. Available at: website www.hafro.is/gadget (accessed 27 April 2012) 95 pp.

Bentley, N., Davies, N.M. and McNeill, S.E. (2004) A spatially explicit model of the snapper (*Pagrus auratus*) fishery in SNA1. New Zealand Fisheries Assessment Report, 1–63.

Blamey, L.K. (2010) *Ecosystem effects of a rock-lobster 'invasion': comparative and modelling approaches*. PhD thesis, University of Cape Town, 256 pp.

Blamey, L.K., Plagányi, É.E. and Branch, G.M. (in press) Modelling a regime shift in a kelp-forest ecosystem caused by a lobster range-expansion. *Bulletin of Marine Science.*

Blanchard, J.L., Coll, M., Trenkel, V.M. *et al.* (2010) Trend analysis of indicators: a comparison of recent changes in the status of marine ecosystems around the world. *ICES Journal of Marine Science: Journal du Conseil* **67**, 732–744.

Blanchard, J., Law, R., Castle, M. and Jennings, S. (2011) Coupled energy pathways and the resilience of size-structured food webs. *Theoretical Ecology* **4**, 289–300.

Bogstad, B., Hauge, K.H. and Ulltang, O. (1997) MULTSPEC – a multispecies model for fish and marine mammals in the Barents Sea. *Journal of Northwest Atlantic Fishery Sciences* **22**, 317–341.

Boncoeur, J., Alban, F., Guyader, O. and Thébaud, O. (2002) Fish, fishers, seals and tourists: economic consequences of creating a marine reserve in a multi-species, multi-activity context. *Natural Resource Modeling* **15**, 387–411.

Bull, B., Francis, R.I.C.C., Dunn, A., McKenzie, A., Gilbert, D.J. and Smith, M.H. (2004) CASAL (C++ algorithmic stock assessment laboratory): CASAL User Manual v2.06e2004/09/26. NIWA Technical Report, 261 pp.

Burford, M.A., Rothlisberg, P.C. and Revill, A.T. (2009) Sources of nutrients driving production in the Gulf of Carpentaria, Australia: a shallow tropical shelf system. *Marine and Freshwater Research* **60**, 1044–1053.

Butterworth, D.S. (2007) Why a management procedure approach? Some positives and negatives. *ICES Journal of Marine Science* **64**, 613.

Butterworth, D.S. and Punt, A.E. (1999) Experiences in the evaluation and implementation of management procedures. *ICES Journal of Marine Science* **56**, 985.

Collie, J.S. and Gislason, H. (2001) Biological reference points for fish stocks in a multispecies context. *Canadian Journal of Fisheries and Aquatic Sciences* **58**, 2167–2176.

Cooke, J.G. (1999) Improvement of fishery-management advice through simulation testing of harvest algorithms. *ICES Journal of Marine Science: Journal du Conseil* **56**, 797–810.

Cury, P.M., Shin, Y.-J., Planque, B. *et al.* (2008) Ecosystem oceanography for global change in fisheries. *Trends in Ecology and Evolution* **23**, 338–346.

Danielsson, A., Stefánsson, G., Baldursson, F.M. and Thorarinsson, K. (1997) Utilization of the Icelandic cod stock in a multispecies context. *Marine Resource Economics* **12**, 329–344.

DeYoung, B., Heath, M., Werner, F., Megrey, B. and Monfray, P. (2004) Challenges of modelling ocean basin ecosystems. *Science* **304**, 1463–1466.

Eppink, F.V. and van den Bergh, J.C.J.M. (2007) Ecological theories and indicators in economic models of biodiversity loss and conservation: a critical review. *Ecological Economics* **61**, 284–293.

Eveson, J.P., Basson, M. and Hobday, A.J. (2012) Using electronic tag data to improve mortality and movement estimates in a tag-based spatial fisheries assessment model. *Canadian Journal of Fisheries and Aquatic Sciences* **69**, 869–883.

FAO (1996) Precautionary approach to capture fisheries and species introductions. Elaborated by the Technical Consultation on the Precautionary Approach to Capture Fisheries (Including Species Introductions). Lysekil, Sweden, 6–13 June 1995. FAO Technical Guidelines for Responsible Fisheries, 1–54.

FAO (2008) Best practices in ecosystem modelling for informing an ecosystem approach to fisheries. *FAO Fisheries Technical Guidelines for Responsible Fisheries*. No. 4, Suppl. 2, Add. 1, Rome, FAO, 1–78.

Flaaten, O. (1998) On the bioeconomics of predator and prey fishing. *Fisheries Research* **37**, 179–191.

Fournier, D. and Archibald, C.P. (1982) A general theory for analyzing catch at age data. *Canadian Journal of Fisheries and Aquatic Sciences* **39**, 1195–1207.

Fournier, D.A., Hampton, J. and Sibert, J.R. (1998) MULTIFAN-CL: a length-based, age-structured model for fisheries stock assessment, with application to South Pacific albacore, *Thunnus alalunga*. *Canadian Journal of Fisheries and Aquatic Sciences* **55**, 2105–2116.

Fournier, D.A., Skaug, H.J., Ancheta, J. *et al.* (2011) AD Model Builder: using automatic differentiation for statistical inference of highly parameterized complex nonlinear models. *Optimization Methods and Software*, **27**, 233–249.

Francis, R.I.C.C. (2011) Data weighting in statistical fisheries stock assessment models. *Canadian Journal of Fisheries and Aquatic Sciences* **68**, 1124–1138.

Fulton, E.A. (2010) Approaches to end-to-end ecosystem models. *Journal of Marine Systems* **81**, 171–183.

Fulton, E.A., Smith, A.D.M. and Johnson, C.R. (2003) Effect of complexity on marine ecosystem models. *Marine Ecology Progress Series* **253**, 1–16.

Fulton, E.A., Fuller, M., Smith, A.D.M. and Punt, A.E. (2004) Ecological indicators of the ecosystem effects of fishing: final report. 116 pp.

Fulton, E.A., Smith, A.D.M., Smith, D.C. and van Putten, I.E. (2011) Human behaviour: the key source of uncertainty in fisheries management. *Fish and Fisheries* **12**, 2–17.

Gaichas, S.K., Aydin, K.Y. and Francis, R.C. (2011) What drives dynamics in the Gulf of Alaska? Integrating hypotheses of species, fishing, and climate relationships using ecosystem modelling. *Canadian Journal of Fisheries and Aquatic Sciences* **68**, 1553–1578.

Garrison, L.P., Link, J.S., Kilduff, D.P. *et al.* (2010) An expansion of the MSVPA approach for quantifying predator–prey interactions in exploited fish communities. *ICES Journal of Marine Science: Journal du Conseil* **67**, 856–870.

Haltuch, M.A. and Punt, A.E. (2011) The promises and pitfalls of including decadal-scale climate forcing of recruitment in groundfish stock assessment. *Canadian Journal of Fisheries and Aquatic Sciences* **68**, 912–926.

Hamon, K., Frusher, S., Little, R. and Thébaud, O. (2010) The effect of quota trading limitation in a fishery managed through individual transferable quotas. (Proceedings of the Conference of the International Institute for Fisheries Economics and Trade, Montpellier, 2010).

Hannesson, R. (1983) Optimal harvesting of ecologically interdependent fish species. *Journal of Environmental Economics and Management* **10**, 329–345.

Hewitt, R.P., Watkins, J., Naganobu, M. *et al.* (2004) Biomass of Antarctic krill in the Scotia Sea in January/February 2000 and its use in revising an estimate of precautionary yield. *Deep Sea Research Part II: Topical Studies in Oceanography* **51**, 1215–1236.

Hill, K., Dorval, E., Lo, N., Macewicz, B., Show, C. and Felix-Uraga, R. (2008) Assessment of the Pacific sardine resource in 2007 for U.S. management in 2008. National Oceanic Atmospheric Administration Technical Memorandum 413 pp.

Hobday, A.J. and Hartmann, K. (2006) Near real-time spatial management based on habitat predictions for a longline bycatch species. *Fisheries Management and Ecology* **13**, 365–380.

Hobday, D., Punt, A.E. and Smith, D.C. (2005) Modelling the effects of Marine Protected Areas (MPAs) on Victoria's southern rock lobster, *Jasus edwardsii*, fishery. *New Zealand Journal of Marine and Freshwater Research* **39**, 675–686.

Hollowed, A.B., Bax, N., Beamish, R. *et al.* (2000) Are multispecies models an improvement on single-species models for measuring fishing impacts on marine ecosystems? *ICES Journal of Marine Science: Journal du Conseil* **57**, 707.

Hollowed, A.B., Aydin, K.Y., Essington, T.E. *et al.* (2011) Experience with quantitative ecosystem assessment tools in the northeast Pacific. *Fish and Fisheries* **12**, 189–208.

Howell, D. and Bogstad, B. (2010) A combined Gadget/FLR model for management strategy evaluations of the Barents Sea fisheries. *ICES Journal of Marine Science: Journal du Conseil* **67**, 1998–2004.

Hunsicker, M.E., Ciannelli, L. and Bailey, K.M. *et al.* (2011) Functional responses and scaling in predator–prey interactions of marine fishes: contemporary issues and emerging concepts. *Ecology Letters*, **14**, 1288–1299.

Hvingel, C. and Kingsley, M.C.S. (2006) A framework to model shrimp (*Pandalus borealis*) stock dynamics and to quantify the risk associated with alternative management options, using Bayesian methods. *ICES Journal of Marine Science: Journal du Conseil* **63**, 68–82.

Imeson, R.J. and van den Bergh, J.C.J.M. (2004) A bioeconomic analysis of a shellfishery: the effects of recruitment and habitat in a metapopulation model. *Environmental and Resource Economics* **27**, 65–86.

IWC (2004) Report of the modelling workshop on Cetacean-Fishery Competition, La Jolla, 25–27 June, 2002. *Journal Cetacean Research and Management* **6**(Suppl), 413–426.

Jurado-Molina, J. and Livingston, P. (2002) Multispecies perspectives on the bering sea groundfish fisheries management regime. *North American Journal of Fisheries Management* **22**, 1164–1175.

Jurado-Molina, J.U., Livingston, P.A. and Ianelli, J.N. (2005) Incorporating predation interactions in a statistical catch-at-age model for a predator-prey system in the eastern Bering Sea. *Canadian Journal of Fisheries and Aquatic Sciences* **62**, 1865–1873.

Kinzey, D. and Punt, A.E. (2009) Multispecies and single-species models of fish population dynamics: comparing parameter estimates. *Natural Resource Modeling* **22**, 67–104.

Koen-Alonso, M. and Yodzis, P. (2005) Multispecies modelling of some components of the marine community of northern and central Patagonia, Argentina. *Canadian Journal of Fisheries and Aquatic Sciences* **62**, 1490–1512.

Lehodey, P., Senina, I. and Murtugudde, R. (2008) A spatial ecosystem and populations dynamics model (SEAPODYM) – modeling of tuna and tuna-like populations. *Progress in Oceanography* **78**, 304–318.

Lindstrøm, U., Smout, S., Howell, D. and Bogstad, B. (2009) Modelling multi-species interactions in the Barents Sea ecosystem with special emphasis on minke whales and their interactions with cod, herring and capelin. *Deep Sea Research Part II: Topical Studies in Oceanography* **56**, 2068–2079.

Little, L.R., Punt, A.E., Mapstone, B.D., Begg, G.A., Goldman, B. and Williams, A.J. (2009) An agent-based model for simulating trading of multi-species fisheries quota. *Ecological Modelling* **220**, 3404–3412.

Livingston, P.A. and Jurado-Molina, J. (2000) A multispecies virtual population analysis of the eastern Bering Sea. *ICES Journal of Marine Science* **57**, 294.

Magnússon, K.G. (1995) An overview of the multispecies VPA – theory and applications. *Reviews in Fish Biology and Fisheries* **5**, 195–212.

Marasco, R.J., Goodman, D., Grimes, C.B. *et al.* (2007) Ecosystem-based fisheries management: some practical suggestions. *Canadian Journal of Fisheries and Aquatic Sciences* **64**, 928–939.

Martinet, V. and Blanchard, F. (2009) Fishery externalities and biodiversity: trade-offs between the viability of shrimp trawling and the conservation of Frigatebirds in French Guiana. *Ecological Economics* **68**, 2960–2968.

Maunder, M.N. (2003) Paradigm shifts in fisheries stock assessment: from integrated analysis to Bayesian analysis and back again. *Natural Resource Modeling* **16**, 465–476.

Maunder, M.N. and Deriso, R.B. (2011) A state–space multistage life cycle model to evaluate population impacts in the presence of density dependence: illustrated with application to delta smelt (*Hyposmesus transpacificus*). *Canadian Journal of Fisheries and Aquatic Sciences* **68**, 1285–1306.

Maunder, M.N. and Watters, G.M. (2003) A general framework for integrating environmental time series into stock assessment models: model description, simulation testing, and example. *Fishery Bulletin* **101**, 89–99.

Maunder, M.N., Starr, P.J. and Hilborn, R. (2000) A Bayesian analysis to estimate loss in squid catch due to the implementation of a sea lion population management plan. *Marine Mammal Science* **16**, 413–426.

Methot, R. (2005) User manual for the assessment program. Stock synthesis II version 1.19. *NOAA Fisheries*, Northwest Fisheries Science Center, 1–47.

Methot, R.D. (2009) Stock assessment: operational models in support of fisheries management. In: *The Future of Fisheries Science in North America*, Vol. 31. (eds R.J. Beamish and B.J. Rothschild). Springer, Dordrecht, Netherlands, pp. 137–165.

Mohn, R. and Bowen, W. (1996) Grey seal predation on the eastern Scotian Shelf: modelling the impact on

Atlantic cod. *Canadian Journal of Fisheries and Aquatic Science* **53**, 2722–2738.

Mori, M. and Butterworth, D.S. (2006) A first step towards modelling the krill – predator dynamics of the Antarctic ecosystem. *CCAMLR Science* **13**, 217–277.

Mullon, C., Mittaine, J.F., Thébaud, O., Péron, G., Merino, G. and Barange, M. (2009) Modeling the global fishmeal and fish oil markets. *Natural Resource Modeling* **22**, 564–609.

Murawski, S.A. (2000) Definitions of overfishing from an ecosystem perspective. *ICES Journal of Marine Science* **57**, 649.

NAFO (2006) Report of the NAFO/ICES *Pandalus* assessment group. NAFO SCS Doc, 1–68.

National Research Council (2006) *Dynamic Changes in Marine Ecosystems: Fishing, Food Webs, and Future Options*. National Academy Press, Washington, DC, pp. 154.

Nostbakken, L., Thebaud, O. and Sorensen, L.-C. (2011) Investment behaviour and capacity adjustment in fisheries: a survey of the literature. *Marine Resource Economics* **26**, 95–117.

NPFMC (2010) *Public Review Draft. Environmental Assessment for proposed Amendments 38 and 29 to the Fishery Management Plan for Bering Sea/Aleutian Islands King and Tanner Crabs to comply with Annual Catch Limit requirements (Amendment 38) and to revise the rebuilding plan for EBS snow crab (Amendment 39)*. North Pacific Fishery Management Council, Anchorage.

Pauly, D., Christensen, V. and Walters, C. (2000) Ecopath, ecosim, and ecospace as tools for evaluating ecosystem impact of fisheries. *ICES Journal of Marine Science* **57**, 697.

Pikitch, E., Boersma, P.D. and Boyd, I.L. *et al.* (2012) Little Fish, Big Impact: Managing a crucial link in ocean food webs. Lenfest Ocean Program. Washington, DC, 108 pp.

Plagányi, É.E. (2007) Models for an ecosystem approach to fisheries. *FAO Fisheries Technical Paper* **477**, 1–126.

Plagányi, É.E. and Butterworth, D.S. (2008) An updated description and parameterisation of the spatial multispecies operating model (SMOM). WG-SAM subgroup of CCAMLR (Commission for the Conservation of Antarctic Marine Living Resources), WG-SAM-08/17, 1–35.

Plagányi, É.E. and Butterworth, D. (2012) The Scotia Sea krill fishery and its possible impacts on dependent predators – modelling localized depletion of prey. *Ecological Applications* **22**, 748–761.

Punt, A.E. and Butterworth, D.S. (1995) The effects of future consumption by the Cape fur seal on catches and catch rates of the Cape hakes. 4. Modelling the biological interaction between Cape fur seals Arctocephalus pusillus pusillus and the Cape hakes Merluccius capensis and M. paradoxus. *South African Journal of Marine Science* **16**, 255–285.

Punt, A.E., Deng, R.A., Dichmont, C.M. *et al.* (2010) Integrating size-structured assessment and bioeconomic management advice in Australia's northern prawn fishery. *ICES Journal of Marine Science: Journal du Conseil* **67**, 1785–1801.

Punt, A.E., Deng, R., Pascoe, S. *et al.* (2011a) Calculating optimal effort and catch trajectories for multiple species modelled using a mix of size-structured, delay-difference and biomass dynamics models. *Fisheries Research* **109**, 201–211.

Punt, A.E., Smith, D.C. and Smith, A.D. (2011b) Among-stock comparisons for improving stock assessments of data-poor stocks: the "Robin Hood" approach. *ICES Journal of Marine Science: Journal du Conseil* **68**, 972.

van Putten, I.E., Kulmala, S. and Thébaud, O. *et al.* (2011) Theories and behavioural drivers underlying fleet dynamics models. *Fish and Fisheries* **12**, 2–17.

Rademeyer, R.A., Plagányi, E.E. and Butterworth, D.S. (2007) Tips and tricks in designing management procedures. *ICES Journal of Marine Science* **64**, 618.

Rayns, N. (2007) The Australian government's harvest strategy policy. *ICES Journal of Marine Science* **64**, 596–598.

Robins, J., Mayer, D., Staunton-Smith, J., Halliday, I., Sawynok, B. and Sellin, M. (2006) Variable growth rates of the tropical estuarine fish barramundi *Lates calcarifer* (Bloch) under different freshwater flow conditions. *Journal of Fish Biology* **69**, 379–391.

Rose, K.A., Allen, J.I., Artioli, Y. *et al.* (2010) End-to-end models for the analysis of marine ecosystems: challenges, issues, and next steps. *Marine and Coastal Fisheries* **2**, 115–130.

Sainsbury, K.J., Punt, A.E. and Smith, A.D.M. (2000) Design of operational management strategies for achieving fishery ecosystem objectives. *ICES Journal of Marine Science* **57**, 731.

Salomon, A.K., Gaichas, S.K., Jensen, O.P. *et al.* (2011) Bridging the divide between fisheries and marine conservation science. *Bulletin of Marine Science* **87**, 251–274.

Sanchirico, J.N. and Wilen, J.E. (2001) A bioeconomic model of marine reserve creation. *Journal of Environmental Economics and Management* **42**, 257–276.

Schirripa, M.J. (2007) *Status of the Sablefish Resource Off the Continental U.S. Pacific Coast in 2007*. Pacific Fishery Management Council, Portland.

Schweder, T., Hagen, G.S. and Hatlebakk, E. (2000) Direct and indirect effects of minke whale abundance on cod and herring fisheries: a scenario experiment for the Greater Barents Sea. *NAMMCO Scientific Publications* **2**, 120–132.

Shertzer, K.W., Prager, M.H. and Williams, E.H. (2008) A probability-based approach to setting annual catch levels. *Fishery Bulletin* **106**, 225–232.

Smith, M. and Wilen, J. (2004) Marine reserves with endogenous ports: empirical bioeconomics of the Cali-

fornia sea urchin fishery. *Marine Resource Economics* **19**, 85–112.

Smith, A.D.M., Sainsbury, K.J. and Stevens, R.A. (1999) Implementing effective fisheries-management systems – management strategy evaluation and the Australian partnership approach. *ICES Journal of Marine Science* **56**, 967–979.

Smith, A.D.M., Fulton, E.J., Hobday, A.J., Smith, D.C. and Shoulder, P. (2007) Scientific tools to support the practical implementation of ecosystem-based fisheries management. *ICES Journal of Marine Science* **64**, 633–639.

Smith, A.D.M., Brown, C.J., Bulman, C.M. *et al.* (2011) Impacts of fishing low–trophic level species on marine ecosystems. *Science* **333**, 1147–1150.

Sparre, P. (1991) Introduction to multispecies virtual population analysis. *ICES Marine Science Symposia* **193**, 12–21.

Stefánsson, G. and Palsson, O.K. (1998) Points of view: a framework for multispecies modelling of Arcto-boreal systems. *Reviews in Fish Biology and Fisheries* **8**, 101–104.

Thorson, J.T. (2011) Auxiliary and focal assessment models: a proof-of-concept involving time-varying catchability and fishery stock status evaluation. *ICES Journal of Marine Science* **68**, 2264–2276.

Tjelmeland, S. and Lindstrøm, U. (2005) An ecosystem element added to the assessment of Norwegian spring-spawning herring: implementing predation by minke whales. *ICES Journal of Marine Science* **62**, 285–294.

Tyrrell, M.C., Link, J.S. and Moustahfid, H. (2011) The importance of including predation in fish population models: implications for biological reference points. *Fisheries Research* **108**, 1–8.

Vermard, Y., Marchal, P., Mahévas, S. and Thébaud, O. (2008) A dynamic model of the Bay of Biscay pelagic fleet simulating fishing trip choice: the response to the closure of the European anchovy (*Engraulis encrasicolus*) fishery in 2005. *Canadian Journal of Fisheries and Aquatic Sciences* **65**, 2444–2453.

Walters, C.J. (1986) *Adaptive Management of Renewable Resources*. MacMillian Publishing Company, New York, pp. 1–570.

Walters, C., Christensen, V. and Pauly, D. (1997) Structuring dynamic models of exploited ecosystems from trophic mass-balance assessments. *Reviews in Fish Biology and Fisheries* **7**, 139–172.

Watters, G.M., Hinke, J.T., Reid, K. and Hill, S. (2005) A krill-predator-fishery model for evaluating candidate management procedures. CCAMLR document WG-EMM-05/13.

Watters, G.M., Hinke, J.T., Reid, K. and Hill, S. (2006) KPFM2, be careful what you ask for – you just might get it. CCAMLR document WG-EMM-06/22.

Watters, G.M., Hinke, J.T. and Hill, S. (2008) A risk assessment to advise on strategies for subdividing a precautionary catch limit among small-scale management units during stage 1 of the staged development of the krill fishery in subareas 48.1, 48.2 and 48.3. CCAMLR document WG-EMM-08/30.

Wilen, J.E., Smith, M.D., Lockwood, D. and Botsford, L.W. (2002) Avoiding surprises: incorporating fisherman behaviour into management models. *Bulletin of Marine Science* **70**, 553–575.

Yodzis, P. (1998) Local trophodynamics and the interaction of marine mammals and fisheries in the Benguela ecosystem. *Journal of Animal Ecology* **67**, 635–658.